# LATHA P

BENGALURU 9398613575

## DETAILS

Bengaluru
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## SKILLS

C Programming

C++

C++ 14

OOPS

STL

Multithreading

Data Structures

GIT

JIRA

SQL

MFC

LINUX

Agile Methodologies

## PROFILE

Detail-oriented and performance-driven software developer with over 4 years of experience in C++ development for embedded systems and enterprise applications. Proven expertise in the automotive domain, particularly in digital instrument cluster and ADAS integration, as well as enterprise financial automation solutions. Skilled in modern C++ (C++14/17), multithreading, STL, CAN communication, and safety-critical software development aligned with ISO 26262 and ASPICE standards. Demonstrated ability to design and implement robust, scalable, and efficient backend systems using structured logic, decision tables, and real-time data processing. Adept at working in cross-functional teams, participating in HIL/SIL testing, and contributing to high-quality software delivery across domains including automotive and financial services.

## EMPLOYMENT HISTORY

**Sr Software Engineer at KPIT TECHNOLOGIES LIMITED, Bengaluru**

April 2022 — April 2025

**Project Name:** Cluster & ADAS Integration
**Client:** Toyota

**Project Description:**

The project focused on the end-to-end development of Toyota's next-generation digital instrument cluster and Advanced Driver Assistance Systems (ADAS) integration, aiming to elevate driver safety, awareness, and user experience. The system provided real-time visualization of vehicle telemetry and ADAS features such as Lane Keep Assist (LKA), Adaptive Cruise Control (ACC), Forward Collision Warning (FCW), and Traffic Sign Recognition (TSR). It involved seamless data exchange with various ECUs over CAN, accurate rendering of safety-critical alerts, and conformance to ISO 26262 functional safety and ASPICE quality standards. Extensive validation through SIL (Software-in-the-Loop) and HIL (Hardware-in-the-Loop) testing ensured robust performance in real-time driving scenarios.

**C++ Roles and Responsibilities:**

- Designed and implemented core cluster functionalities using **Modern C++ (C++14/17)** on embedded platforms.
- Developed **multithreaded C++ modules** for parallel processing of sensor data, rendering tasks, and CAN message handling with thread safety mechanisms (mutexes, condition variables).
- Utilized **Standard Template Library (STL)** containers and algorithms for efficient data processing and management.
- Integrated and parsed **CAN messages** from multiple vehicle ECUs to update real-time displays and warning indicators.
- Built reusable software components for displaying vehicle speed, navigation prompts, driver warnings, and ADAS feedback in the digital cluster.
- Mapped backend C++ logic to UI elements, working closely with HMI teams to ensure real-time rendering and compliance with Toyota's design guidelines.
- Followed **MISRA C++ guidelines**, performed static code analysis, and participated in code reviews to maintain code safety and reliability.
- Developed **unit tests and test harnesses** for validating C++ modules; supported automated regression testing pipelines.
- Conducted SIL and HIL validation to verify the system's real-time responsiveness and accuracy in dynamic automotive scenarios.
- Collaborated in hazard analysis and ensured compliance with **ISO 26262** for safety-critical feature delivery.

**Software Engineer at DIEBOLD NIXDORF INDIA PRIVATE LIMITED, Hyderabad**
June 2021 — February 2022

**Project Name:** Auto Cash Replenishment System

**Project Description:**
This project aimed to automate the cash application process for companies receiving high volumes of customer payments, reducing manual workload and minimizing delays in deposit processing. The solution was built around an intelligent **Auto Cash Replenishment Application** that leveraged structured decision tables and rule-based matching algorithms to automatically reconcile incoming payments with open accounts receivable. The application ingested payment data—including MICR (Magnetic Ink Character Recognition) information and transaction amounts—from a lockbox data feed and processed it through a multi-step decision framework. The system ensured high-accuracy matching by first linking bank account numbers to customer profiles, followed by invoice-level matching based on exact payment equivalency. Any unmatched or ambiguous transactions were escalated for manual intervention, thus improving overall processing efficiency while maintaining accuracy and auditability.

**C++ Roles and Responsibilities:**

- Developed **backend processing logic in C++**, implementing business rules for automated matching of customer payments to open receivables using a structured decision table.
- Used **STL containers and algorithms** to efficiently manage and search large datasets including invoice records, customer mappings, and transaction histories.
- Designed and implemented **multithreaded modules** to handle concurrent processing of lockbox data streams and customer record access, ensuring performance and data integrity.
- Integrated payment processing components with external **data feed systems** using file parsers and stream-based input mechanisms for MICR and transaction data.
- Developed **modular, reusable C++ classes** for invoice matching logic, rule evaluation, and result categorization (auto-match vs. manual-review buckets).
- Applied **decision table logic** to evaluate and apply payments based on configurable rules such as exact match, multiple invoice match, and fallback to manual review.
- Wrote **unit tests and validation tools** for decision logic and transaction processing modules to ensure correctness across edge cases.
- Participated in **code reviews and static analysis** for quality assurance and performance optimizations.

## EDUCATION

**MCA, Chadalavada Ramanamma Engineering College, Tirupati**
June 2015 — April 2018

**B.SC[Statistics], Sri Venkateswara PG & Degree college, Anantapur**
June 2012 — April 2015

**Intermediate (MPC), Sri Vivekananda Jr College, Anantapur**
June 2010 — April 2012

**S.S.C, Z.P.High School, Anantapur**
June 2009 — March 2010